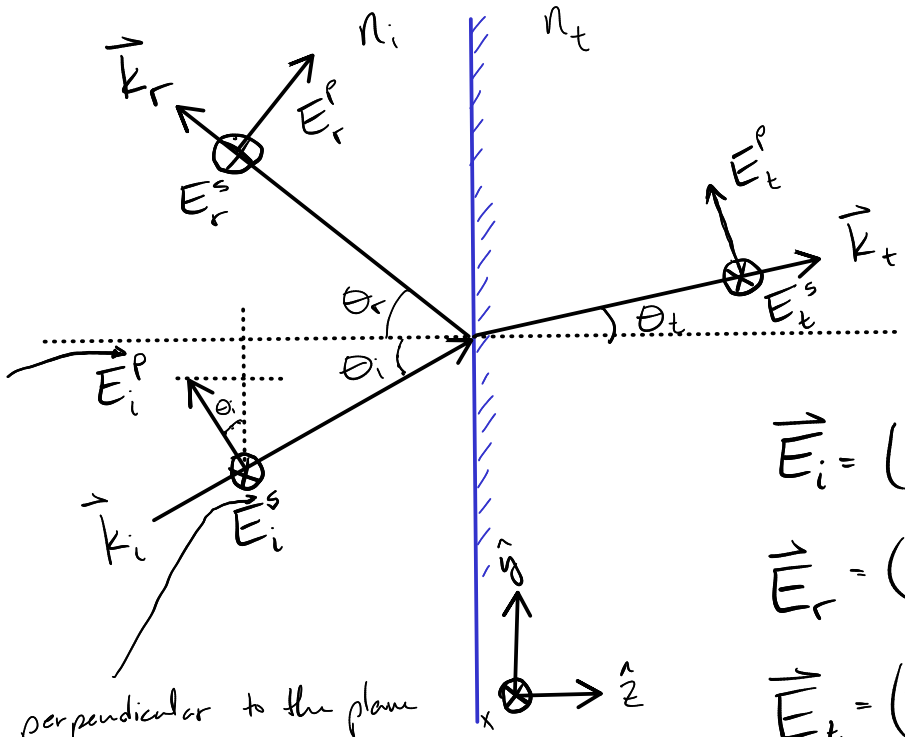
Chapter 3 - Reflection + Refraction

260316 M



$$\vec{k}_i = k_i(\hat{y}\sin\theta_i + \hat{z}\cos\theta_i)$$

$$\vec{k}_r = k_r(\hat{y}\sin\theta_r - \hat{z}\cos\theta_r)$$

$$\vec{k}_t = k_t(\hat{y}\sin\theta_t + \hat{z}\cos\theta_t)$$

"parallel" any vector in the plane is parallel to the plane

perpendicular to the plane "senkrecht"

$$\vec{E}_i = \left(E_i^p(\hat{y}\cos\theta_i - \hat{z}\sin\theta_i) + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)}$$

$$\vec{E}_r = \left(E_r^p(\hat{y}\cos\theta_r + \hat{z}\sin\theta_r) + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$\vec{E}_t = \left(E_t^p(\hat{y}\cos\theta_t - \hat{z}\sin\theta_t) + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

from Maxwell's

↳ Boundary conditions require the components parallel to the interface to be continuous: ($\hat{x}$ + $\hat{y}$ directions in this example)

$$\left(E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)} + \left(E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$= \left(E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

$\omega_i = \omega_r = \omega_t = \omega$

$$\underbrace{k_i \sin\theta_i = k_r \sin\theta_r}$$ $= k_t \sin\theta_t$

$k = \frac{\omega}{v}$

$v = \frac{c}{n}$

$k_i = \frac{\omega}{v} = \frac{n_i \omega}{c}$ $\quad k_r = \frac{\omega}{v} = \frac{n_i \omega}{c}$

$k_i = k_r$

$\sin\theta_i = \sin\theta_r$

$\boxed{\theta_i = \theta_r}$ law of reflection

$k_i \sin\theta_i = k_t \sin\theta_t$

$\frac{n_i \cancel{\omega}}{\cancel{c}} \sin\theta_i = \frac{n_t \cancel{\omega}}{\cancel{c}} \cdot \sin\theta_t$

$\boxed{n_i \sin\theta_i = n_t \sin\theta_t}$ Snell's Law

$$E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s + E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s = E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s$$

x-direction

$$(1)\quad E_i^s + E_r^s = E_t^s$$

y-direction

$$E_i^p\cos\theta_i + E_r^p\cos\theta_r = E_t^p\cos\theta_t$$

$$\theta_i = \theta_r$$

$$(2)\quad \left(E_i^p + E_r^p\right)\cos\theta_i = E_t^p\cos\theta_t$$

So we have 2 equations but 4 unknowns, so we need the magnetic field equations.

$$\vec{B} = \frac{\vec{k}\times\vec{E}}{\omega} = \frac{n}{c}\,\hat{u}\times\vec{E} \quad \text{where} \quad \hat{u} = \frac{\vec{k}}{|k|}$$

$$(3)\quad n_i\left(E_i^s - E_r^s\right)\cos\theta_i = n_t E_t^s\cos\theta_t$$

$$(4)\quad n_i\left(E_i^p - E_r^p\right) = n_t E_t^p$$

Fresnel Equations

$$r_s = \frac{E_r^s}{E_i^s}$$

$$t_s = \frac{E_t^s}{E_i^s}$$

$$E_i^s + E_r^s = E_t^s \quad \Big| \quad \frac{\cancel{n_i \cos\theta_i}\left(E_i^s - E_r^s\right)}{\cancel{n_i \cos\theta_i}} = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

solve w/ elimination

$$E_i^s - E_r^s = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

$$2E_i^s = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i} + 1\right) E_t^s$$

$$2E_i^s = \frac{n_t \cos\theta_t + n_i \cos\theta_i}{n_i \cos\theta_i} \cdot E_t^s$$

$$\boxed{t_s = \frac{E_t^s}{E_i^s} = \frac{2 n_i \cos\theta_i}{n_t \cos\theta_t + n_i \cos\theta_i}}$$

$$\boxed{t_p = \frac{E_t^p}{E_i^p} = \frac{2 n_i \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

$$\boxed{r_s = \frac{E_r^s}{E_i^s} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}}$$

$$\boxed{r_p = \frac{E_r^p}{E_i^p} = \frac{n_i \cos\theta_t - n_t \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

recall:

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \sin\theta_i\right)$$

Reflectance + Transmittance

260320 F

Conservation of energy: $P_i = P_r + P_t$ ← power

$$P_i^s = P_r^s + P_t^s$$

$$P_i^p = P_r^p + P_t^p$$

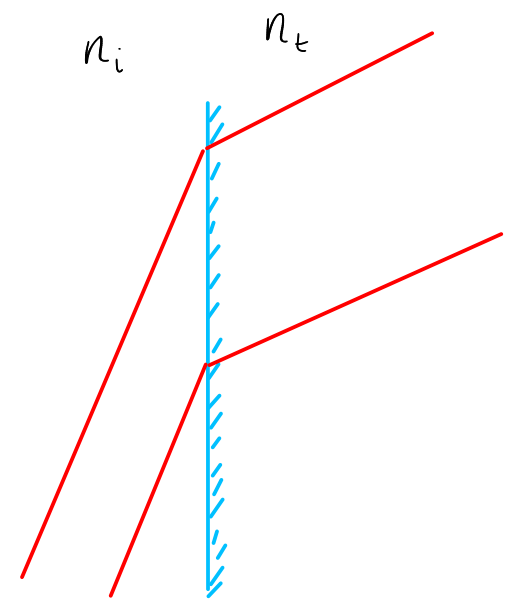
irradiance (intensity)

$$I = \frac{P}{A} \qquad I = c^2 \epsilon_0 \langle E^2 \rangle$$

$$P \propto I \propto E^2$$ ← electric field

reflectance

$$R_s = \frac{P_r^s}{P_i^s} = |r_s|^2 \qquad R_p = \frac{P_r^p}{P_i^p} = |r_p|^2$$



conservation of energy

$$R + T = 1 \Rightarrow \underline{\underline{T = 1 - R}}$$

$$T \neq t^2$$

$$T_s = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_s|^2 \qquad T_p = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_p|^2$$

Brewster's angle

$$\theta_i + \theta_t = \frac{\pi}{2}$$

Snell's $\rightarrow$ $n_i \sin\theta_i = n_t \sin\theta_t$

$$n_i \sin\theta_i = n_t \underbrace{\sin\left(\frac{\pi}{2} - \theta_i\right)}_{\cos\theta_i}$$ $\leftarrow$ at one particular place

$$n_i \sin\theta_B = n_t \cos\theta_B$$ $\leftarrow$ Brewster's angle

$$\tan\theta_B = \frac{n_t}{n_i} \Rightarrow \theta_B = \tan^{-1}\left(\frac{n_t}{n_i}\right)$$